

BANK LOAN CASE STUDY

PROJECT DESCRIPTION: The case study aims to identify a pattern if a client has difficulty paying their installment. This project also provides an example of how EDA (EXPLORATORY DATA ANALYSIS) might be used in the real-world corporate setting. Some learning about risk analytics in banking. The goal is to detect what are the trends that show whether a client has trouble paying their installments. The process starts from data cleaning to the basic method of data analysis such as univariate, segment univariate and bivariate analysis and correlation between variables

APPROACH: To perform the project a systematic approach was followed. A dataset was downloaded. Microsoft Excel 2021 was selected as the primary tool for data analysis such as cleaning, missing data and other EDAs as needed due to its versatility and robust capabilities in handling tabular data. Specific techniques such as pivot tables, charts, and formulas were employed to analyze the dataset and extract meaningful insights.

TECH-STACK USED: Software: Microsoft Excel 2021

Purpose: Excel was chosen for its extensive data analysis functions, including pivot tables, charts, and statistical functions and visualization of the hiring data

INSIGHTS: various key insights were uncovered through the data analytics process. Meaningful trends and visualization patterns were observed in the data

A. IDENTIFY MISSING DATA AND DEAL WITH IT APPROPRIATELY: As a data analyst, you come across missing data in the loan application dataset. It is essential to handle missing data effectively to ensure the accuracy of the analysis.

Task: Identify the missing data in the dataset and decide on an appropriate method to deal with it using Excel built-in functions and features.

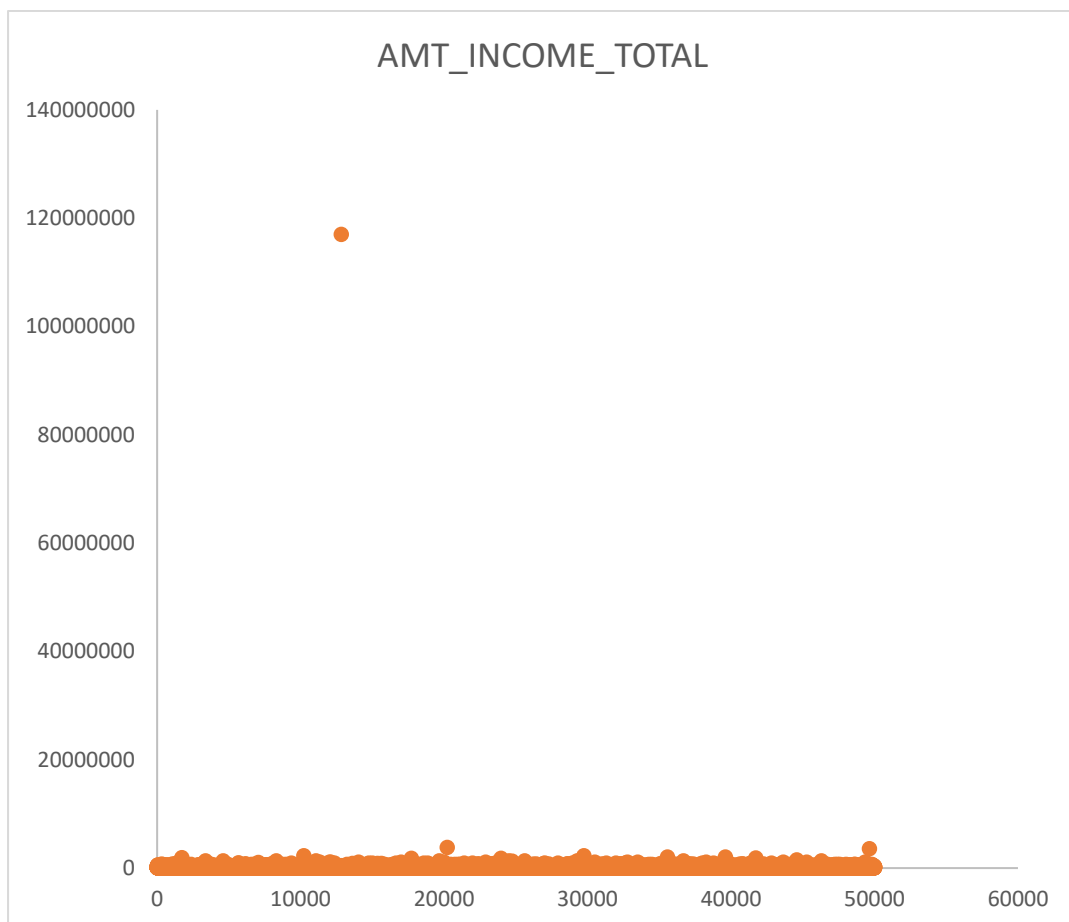
PROPORTION MISSING VALUE	MODIFIED METHOD
< 50%	FOR NUMERICAL VALUES AVERAGE AND MEDIAN WILL BE FOR CATEGORICAL VALUES MODE WILL BE PREFERRED
>50	REMOVE THE WHOLE COLUMN COMPLETELY

SL.NO	REPLACING MISSING COLUMNS	FORMULA(AVERAGE)	MEDIAN
1	AVERAGE AMT_ ANNUITY	27107.38	24939
2	AVERAGE AMT_ GOODS_ PRICE	539060.04	450000
3	NAME_ TYPE_ SUITE	Unaccompanied	MODE
4	OCCUPATION_ TYPE	Laborers	MODE
5	AVERAGE CNT_ FAM_ MEMBERS	2.16	2
6	OBS_ 30_ CNT_ SOCIAL_ CIRCLE	1.42	0
7	DEF_ 30_ CNT_ SOCIAL_ CIRCLE	0.14	0
8	OBS_ 60_ CNT_ SOCIAL_ CIRCLE	1.40	0
9	DEF_ 60_ CNT_ SOCIAL_ CIRCLE	0.10	0
10	DAYS_ LAST_ PHONE_ CHANGE	-964	-755
11	AMT_ REQ_ CREDIT_ BUREAU_ HOUR	0	0
12	AMT_ REQ_ CREDIT_ BUREAU_ DAY	0	0
13	AMT_ REQ_ CREDIT_ BUREAU_ WEEK	0	0
14	AMT_ REQ_ CREDIT_ BUREAU_ MON	0	0
15	AMT_ REQ_ CREDIT_ BUREAU_ QRT	0	0
16	AMT_ REQ_ CREDIT_ BUREAU_ YEAR	2	2

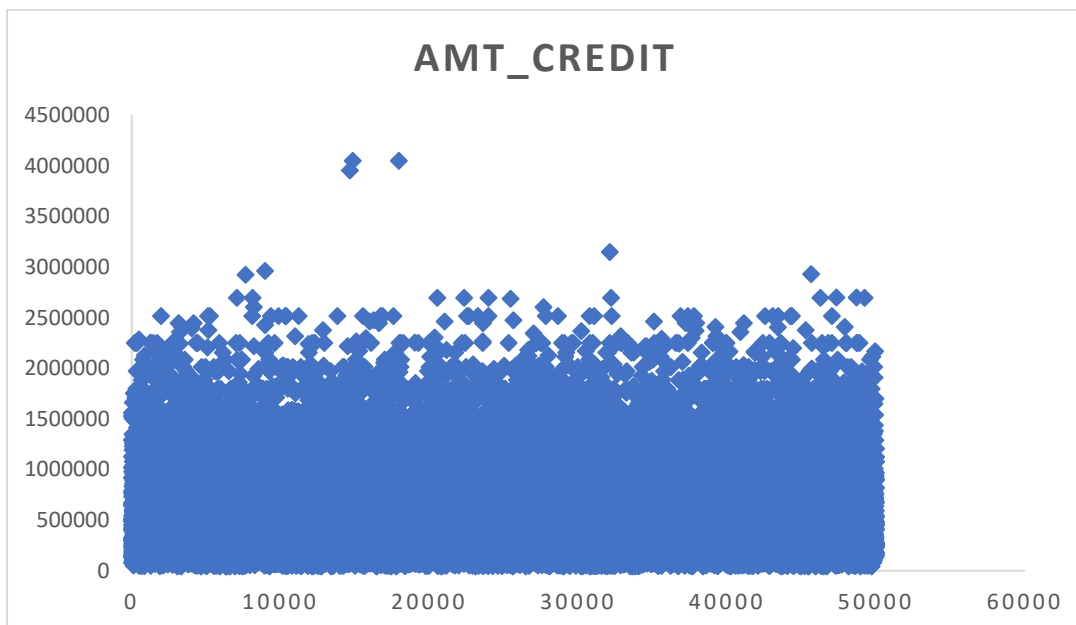
B. Identify Outliers in the Dataset: Outliers can significantly impact the analysis and distort the results. You need to identify outliers in the loan application dataset.

Task: Detect and identify outliers in the dataset using Excel statistical functions and features, focusing on numerical variables.

Q1	112500
Q3	202500
IQR(Q3-Q1)	90000
MIN	-22500
MAX	225000
MEDIAN	145800



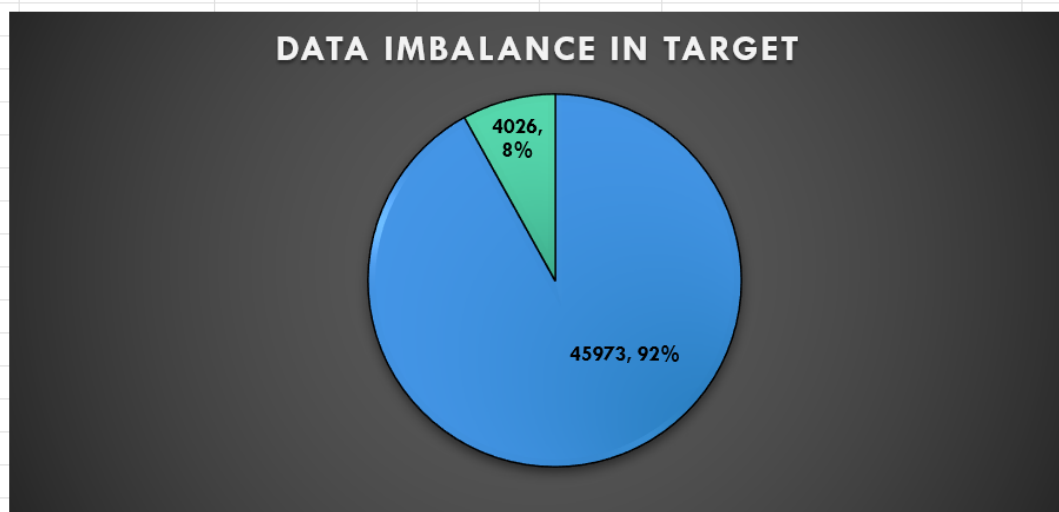
Q1	270000
Q3	808650
IQR(Q3-Q1)	538650
MIN	-537975
MAX	1616625
MEDIAN	514777.5



C. Analyse Data Imbalance: Data imbalance can affect the accuracy of the analysis, especially for binary classification problems. Understanding the data distribution is crucial for building reliable models.

Task: Determine if there is data imbalance in the loan application dataset and calculate the ratio of data imbalance using Excel functions.

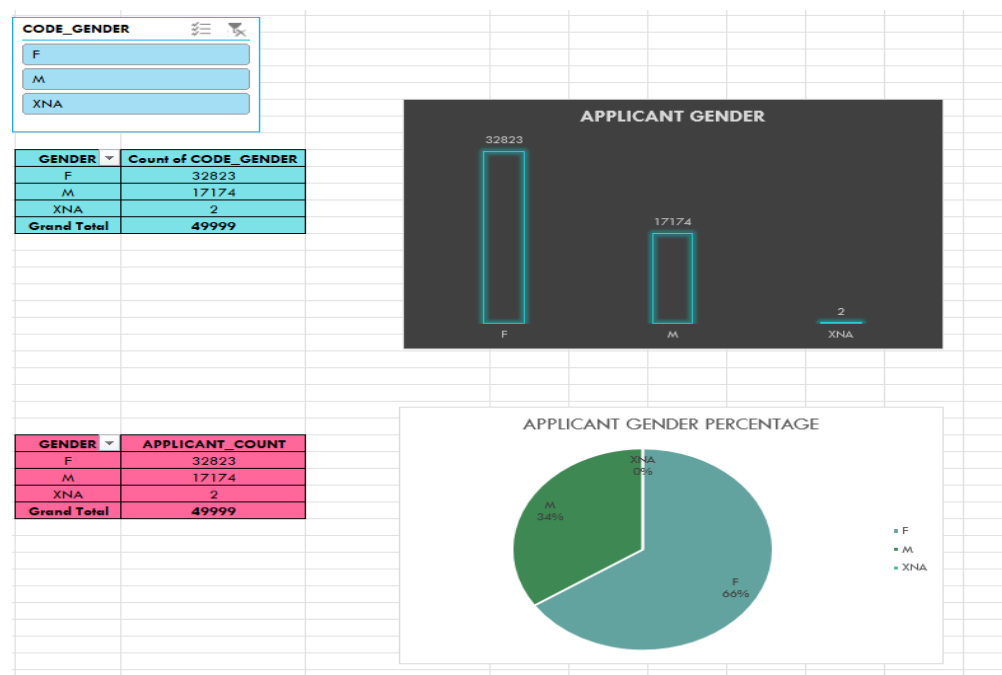
TARGET	Count of TARGET	TARGET	
0	45973	0 NO PAYMENT DIFFICULTY	45973
1	4026	1 DEFAULT CHANCE	4026
Grand Total	49999	RATIO OF DATA IMBALANCE	11.42

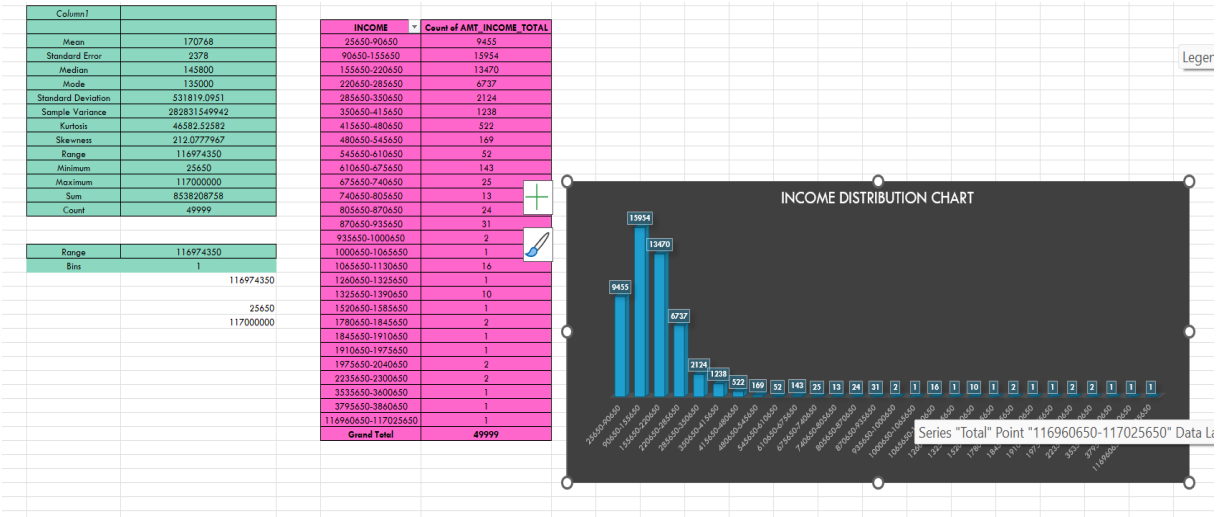
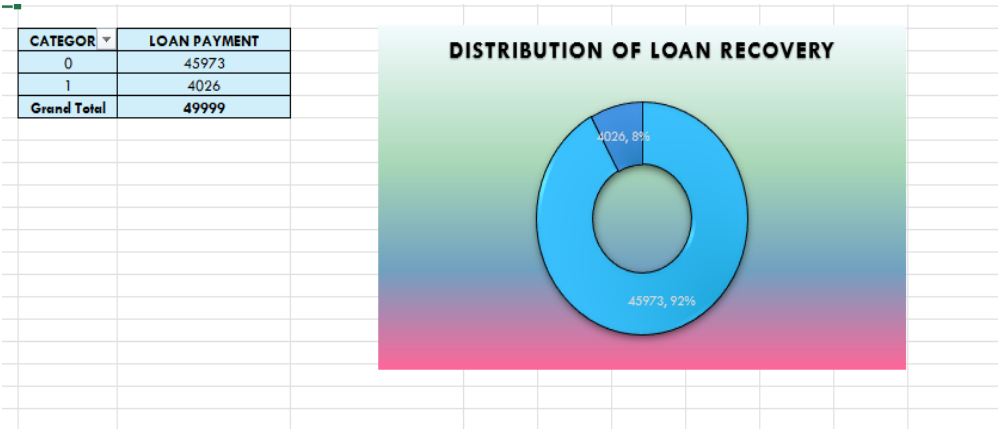


D. Perform Univariate, Segmented Univariate, and Bivariate Analysis: To gain insights into the driving factors of loan default, it is important to conduct various analyses on consumer and loan attributes.

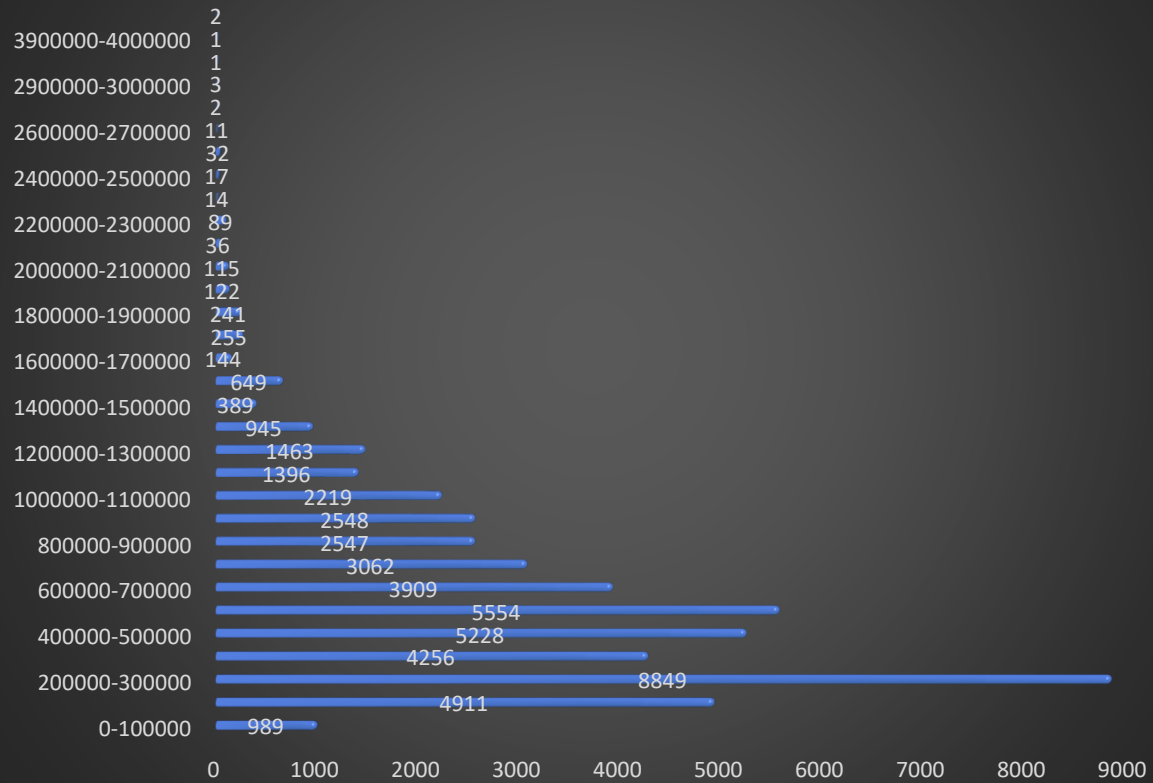
Task: Perform univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable using Excel functions and features.

UNIVARIATE ANALYSIS:

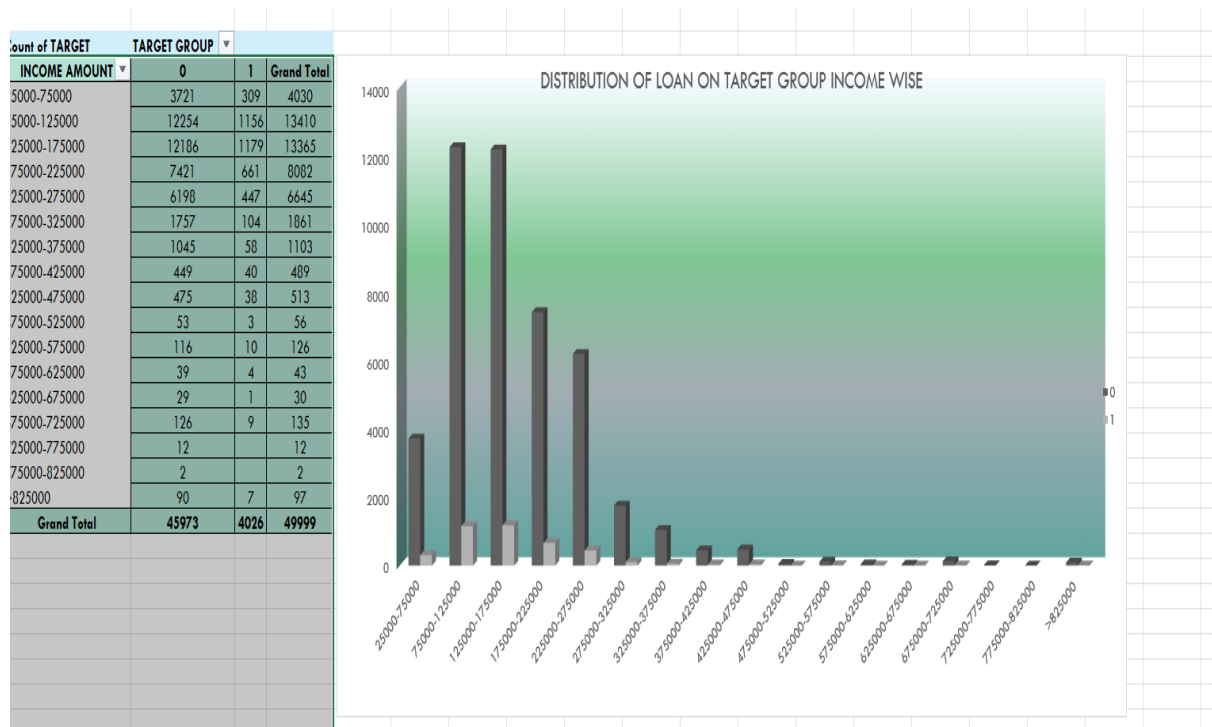




CREDIT DISTRIBUTION



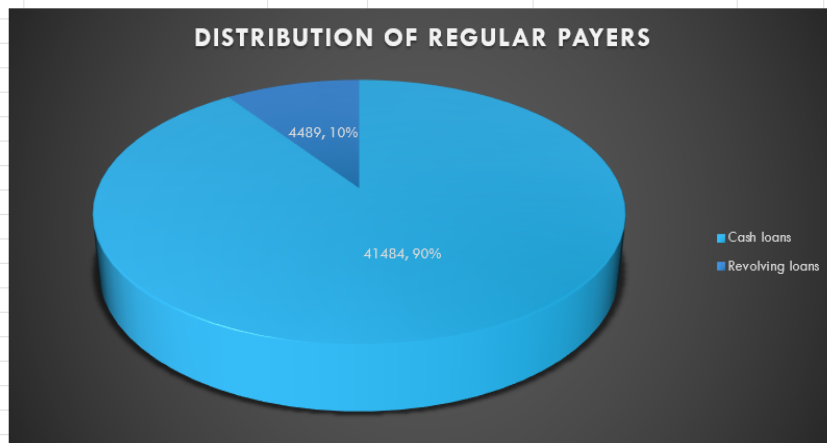
SEGMENT UNIVARIATE ANALYSIS:

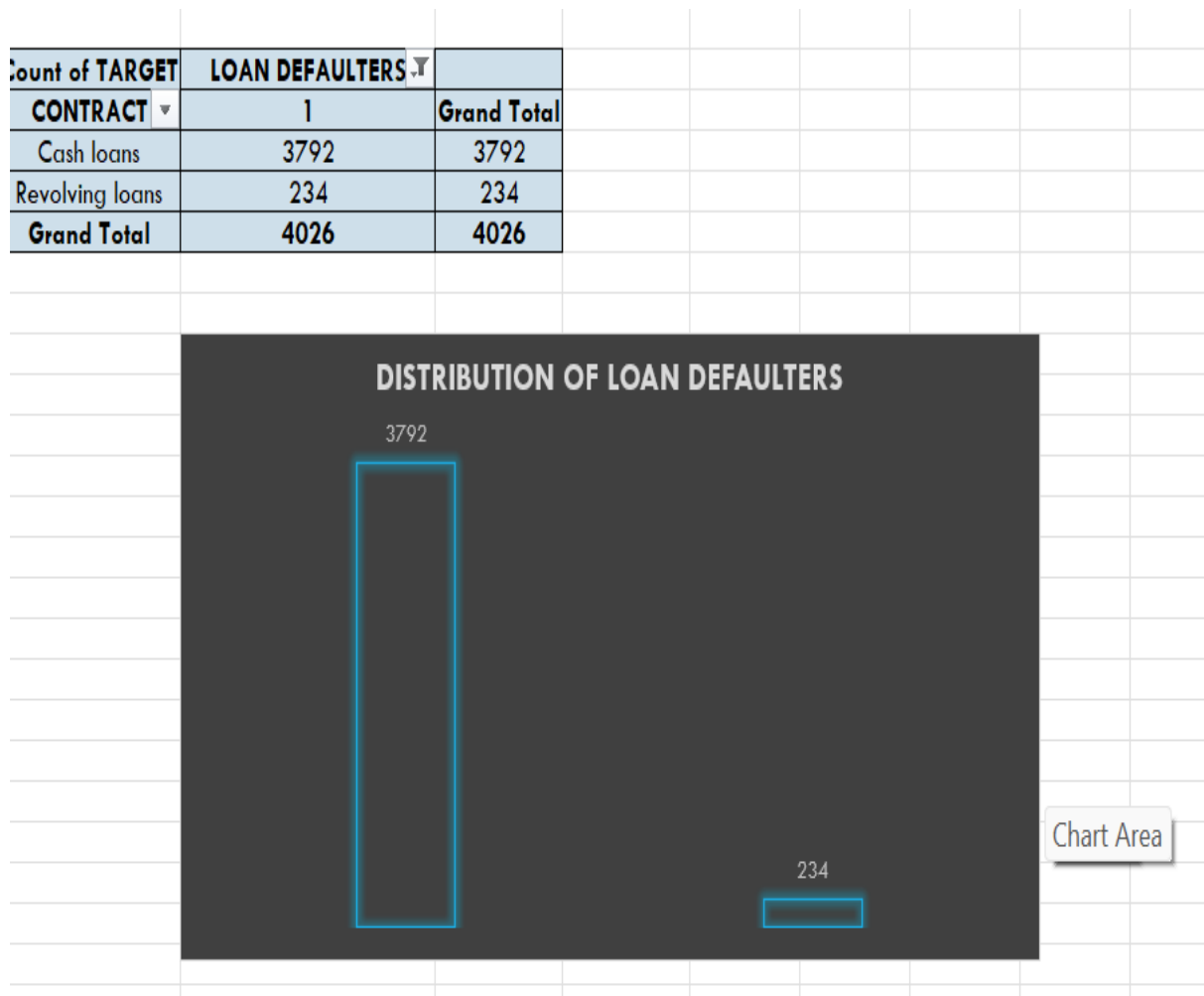


Count of TARGET	LOAN REGULAR PAYERS	
CONTRACY TYPE	0	Grand Total
Cash loans	41484	41484
Revolving loans	4489	4489
Grand Total	45973	45973

Count of TARGET	LOAN REGULAR PAYERS
CONTRACY TYPE	0
Cash loans	41484
Revolving loans	4489
Grand Total	45973

0 LOAN REGU
1 DEFAULTERS

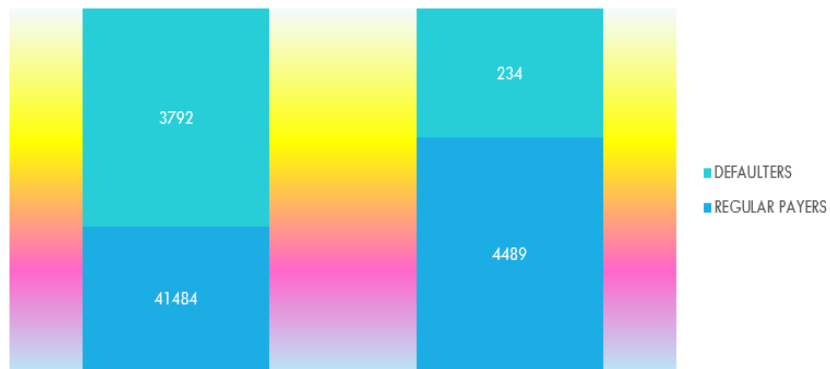




BIVARIATE ANALYSIS:

A	B	C	D	E	F	G	H	I
Count of TARGET	TARGET GROUP							
CONTRACT TYPE	REGULAR PAYERS	DEFAULTERS	Grand Total		CONTRACT TYPE	REGULAR PAYERS	DEFAULTERS	
Cash loans	41484	3792	45276		Cash loans	41484	3792	
Revolving loans	4489	234	4723		Revolving loans	4489	234	
Grand Total	45973	4026	49999		Grand Total	45973	4026	

LOAN CATEGORY AGAINST TARGET

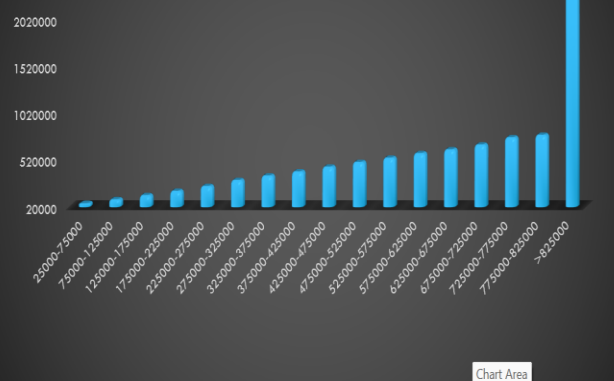


0 REGULAR PAYERS

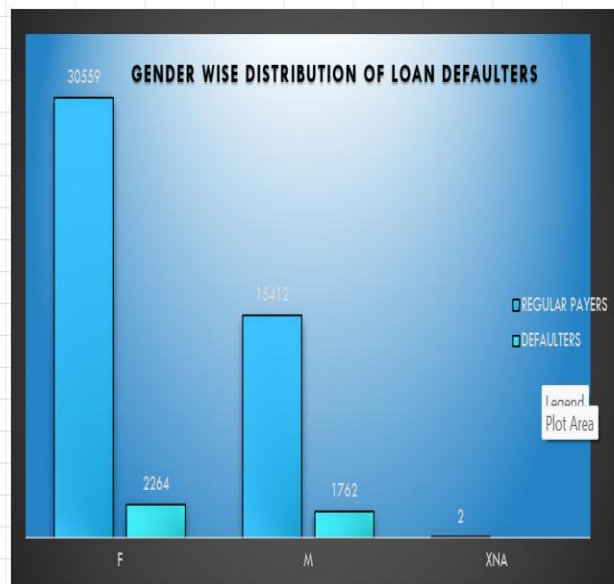
1 DEFAULTERS

INCOME AMOUNT	Average of AMT. INCOME TOTAL
25000-75000	60893.22
75000-125000	101124.41
125000-175000	145135.07
175000-225000	190600.10
225000-275000	241976.08
275000-325000	305219.24
325000-375000	352801.16
375000-425000	396784.97
425000-475000	447067.15
475000-525000	496896.43
525000-575000	541946.43
575000-625000	590860.47
625000-675000	630750.00
675000-725000	681650.00
725000-775000	758473.13
775000-825000	785250.00
>825000	2319355.67
Grand Total	170767.59

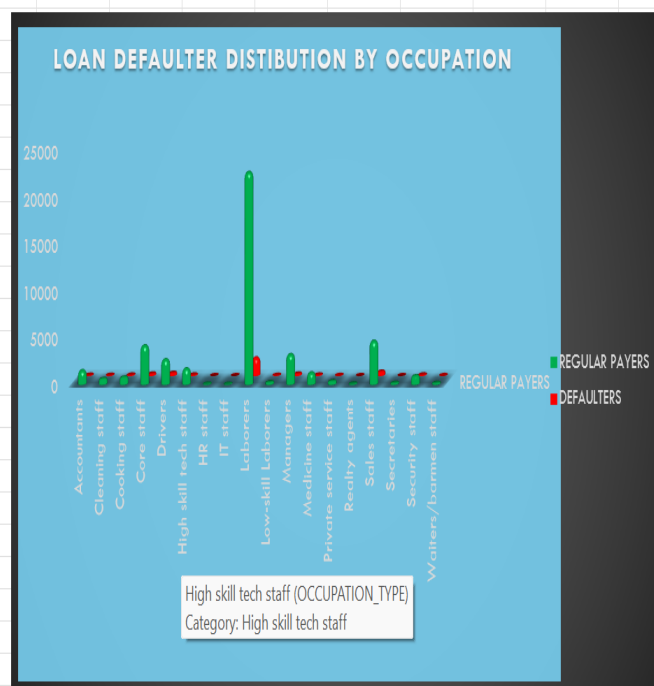
AVERAGE INCOME AMOUNT



Count of TARGET	TARGET GROUP		
GENDER	REGULAR PAYERS	DEFAULTERS	Grand Total
F	30559	2264	32823
M	15412	1762	17174
XNA	2		2
Grand Total	45973	4026	49999



Count of TARGET	TARGET GROUP		
OCCUPATION	REGULAR PAYERS	DEFAULTERS	Grand Total
Accountants	1540	81	1621
Cleaning staff	671	68	739
Cooking staff	862	101	963
Core staff	4184	250	4434
Drivers	2706	338	3044
High skill tech staff	1734	118	1852
HR staff	92	9	101
IT staff	76	4	80
Laborers	22660	1946	24606
Low-skill Laborers	296	61	357
Managers	3246	243	3489
Medicine staff	1297	106	1403
Private service staff	410	37	447
Realty agents	110	13	123
Sales staff	4668	492	5160
Secretaries	203	9	212
Security staff	1015	125	1140
Waiters/barmen staff	203	25	228
Grand Total	45973	4026	49999



E. Identify Top Correlations for Different Scenarios: Understanding the correlation between variables and the target variable can provide insights into strong indicators of loan default.

Task: Segment the dataset based on different scenarios (e.g., clients with payment difficulties and all other cases) and identify the top correlations for each segmented data using Excel functions.

TARGET (0):

CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_ID_PUBLISH	REGION_RATING_CLIENT	AMT_REQ_CREDIT_BUREAU_YEAR
1.00	0.04	0.01	0.00	-0.02	-0.34	-0.25	0.03	0.02	-0.04
0.04	1.00	0.38	0.38	0.18	-0.07	-0.16	-0.03	-0.21	0.02
0.01	0.38	1.00	0.99	0.10	0.05	-0.07	0.01	-0.10	-0.04
0.00	0.38	0.99	1.00	0.10	0.05	-0.07	0.01	-0.10	-0.05
-0.02	0.18	0.10	0.10	1.00	0.03	-0.01	0.00	-0.54	0.01
-0.34	-0.07	0.05	0.05	0.03	1.00	0.62	0.27	-0.01	0.06
-0.25	-0.16	-0.07	-0.07	-0.01	0.62	1.00	0.27	0.04	0.05
0.03	-0.03	0.01	0.01	0.00	0.27	0.27	1.00	0.01	0.03
0.02	-0.21	-0.10	-0.10	-0.54	-0.01	0.04	0.01	1.00	0.01
-0.04	0.02	-0.04	-0.05	0.01	0.06	0.05	0.03	0.01	1.00

TARGET (1):

AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_ID_PUBLISH	REGION_RATING_CLIENT	AMT_REQ_CREDIT_BUREAU_YEAR
0.01	0.01	0.00	-0.02	-0.25	-0.19	0.04	0.06	-0.04
1.00	0.02	0.01	-0.01	-0.01	-0.01	0.01	-0.01	-0.01
0.02	1.00	0.98	0.07	0.14	0.02	0.04	-0.05	-0.03
0.01	0.98	1.00	0.08	-0.14	0.14	0.02	0.02	-0.05
-0.01	0.07	0.08	1.00	0.02	0.01	0.01	-0.43	0.02
-0.01	0.14	0.14	0.02	1.00	0.59	0.25	-0.05	0.09
-0.01	0.02	0.02	0.01	0.59	1.00	0.23	-0.01	0.02
0.01	0.04	0.05	0.01	0.25	0.23	1.00	-0.03	0.06
-0.01	-0.05	-0.05	-0.43	-0.05	-0.01	-0.03	1.00	0.02
-0.01	-0.03	-0.03	0.02	0.09	0.02	0.06	0.02	1.00

TOP 5 CORRELATION (TAGE 0)- LOAN PAYERS		
VARIABLE 1	VARIABLE 2	CORRELATION
DAYS_EMPLOYED	DAYS_BIRTH	0.62
AMT_CREDIT	AMT_INCOME_TOTAL	0.38
AMT_GOODS_PRICE	AMT_INCOME_TOTAL	0.27
DAYS_ID_PUBLISH	DAYS_BIRTH	0.27
DAYS_ID_PUBLISH	DAYS_EMPLOYED	0.27

TOP 5 CORRELATION (TARGET 1)- DEFAULTERS		
VARIABLE 1	VARIABLE 2	CORRELATION
AMT_GOODS_PRICE	AMT_CREDIT	0.98
DAYS_BIRTH	DAYS_EMPLOYED	0.59
DAYS_ID_PUBLISH	DAYS_BIRTH	0.25
DAYS_ID_PUBLISH	DAYS_EMPLOYED	0.23
DAYS_BIRTH	AMT_GOODS_PRICE	0.14

RESULTS: The Bank Lona case study was executed starting with data cleaning process as missing data, identifying outliers and other exploratory data analysis then followed by different analysis and then identify the correlation between different variables

Excel link for file: [PROJECT-6-BANK LOAN ANALYSIS.xlsx](#)

PPT VIDEO LINK: https://drive.google.com/file/d/1s4_Q9pWEpoGOO-ojlr6aCb1NFm-B7NZu/view?usp=sharing



THANK
YOU